

EEE F313: ADVD Lab 3 Report

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1 Introduction

This lab had the goal of performing both static and transient analysis on a CMOS inverter. The CMOS inverter circuit is shown below in Figure 1.

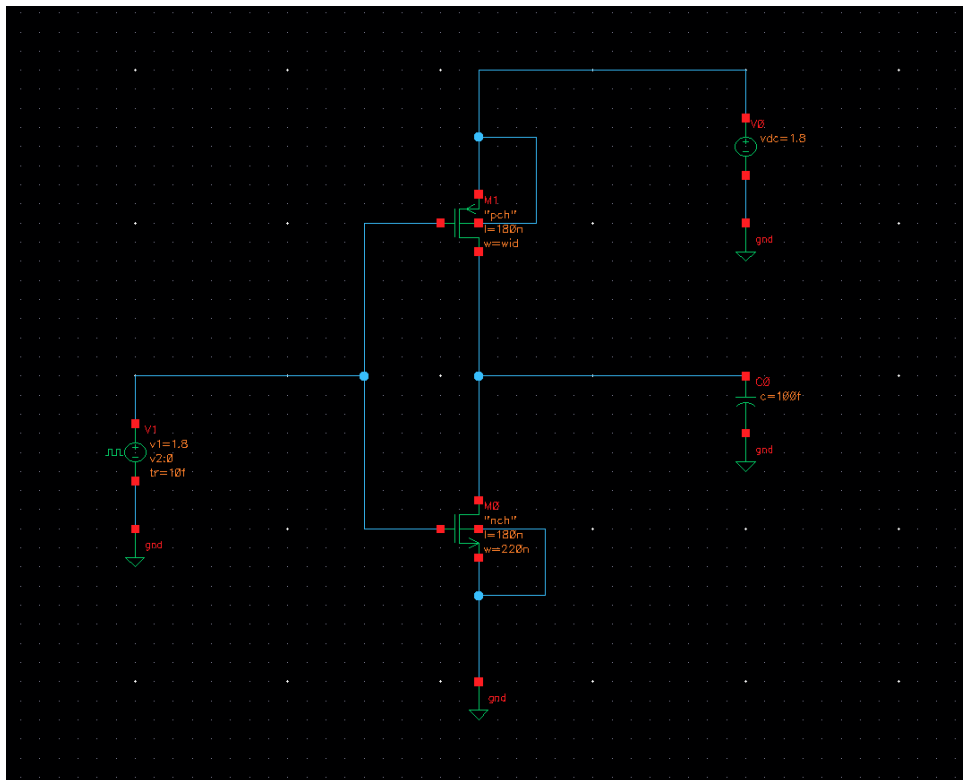


Figure 1: CMOS Inverter Circuit

2 Identifying CMOS Inverter Parameters

From the CMOS circuit in Figure 1, the following parameters were identified by plotting the VTC curve and it's derivative in Cadence. Both the curves are shown below in Figures 2a and 2b respectively.

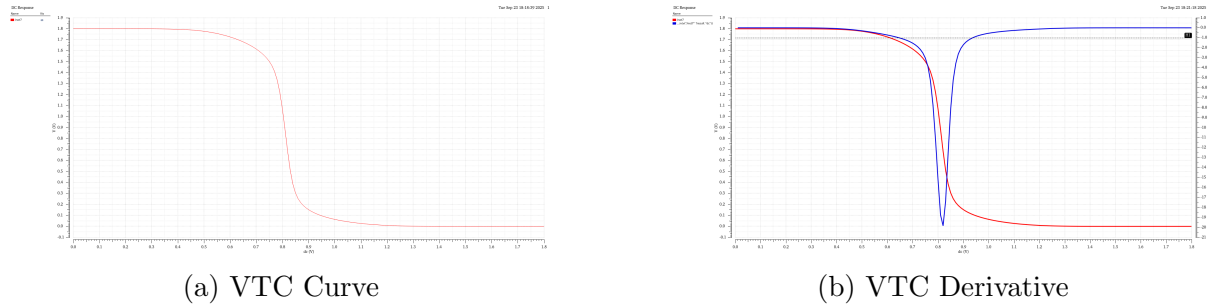


Figure 2: VTC Curves of the Basic CMOS Circuit

From the VTC curve, the following parameters were identified:

- $V_{OH} = 1.8V = V_{DD}$
- $V_{OL} = 0.2030279 \text{ mV} \neq \text{GND}$
- $V_{IL} = 0.49974 \text{ V}$
- $V_{IH} = 1.19555 \text{ V}$

3 Designing a Symmetric CMOS Inverter

This next section of the lab required the design of the CMOS inverter to be symmetric. This was done by adjusting the width of the PMOS transistor from 660 nm to 1460nm. A straight line of $V_{out} = V_{in}$ was plotted on the VTC curve to identify the width at which the PMOS VTC intersected this line. Upon the first iteration of the parametric analysis shown in Figure 3a, it was determined that the width of the PMOS should be between 1160nm and 1260nm. Figure 3b shows this portion of the graph zoomed in.

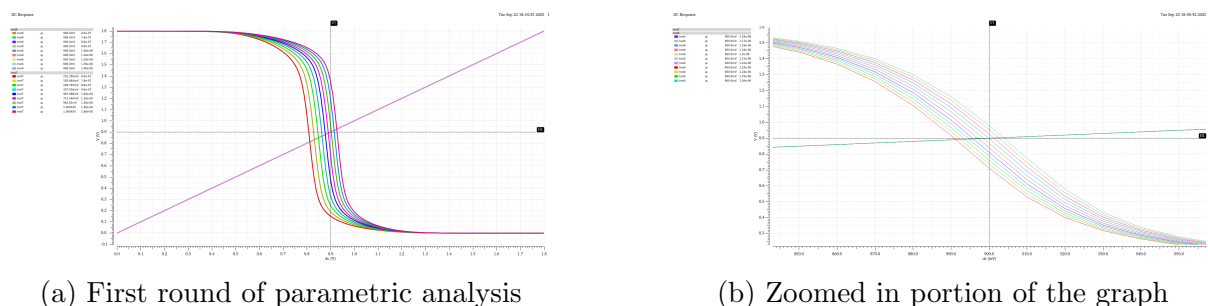


Figure 3: Parametric analysis of PMOS width round 1

Now, upon a second round of parametric analysis between the widths of 1160nm and 1260nm, the following graph in Figure 4 was obtained.

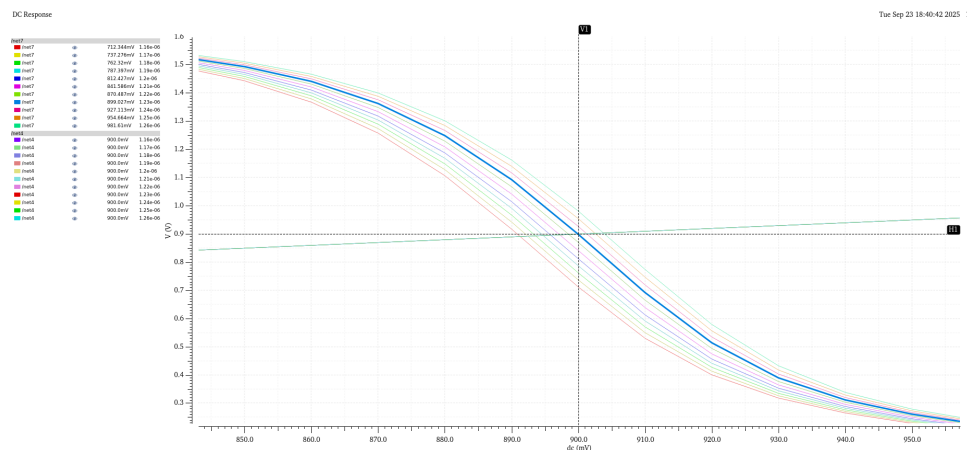


Figure 4: Second round of parametric analysis

This graph shows that the width of the PMOS should be 1230nm to obtain a symmetric CMOS inverter for this NMOS configuration. For the next part of this lab, that particular value was used.

4 Transient Analysis of the Symmetric CMOS Inverter

The final part of this lab involved performing a transient analysis of the symmetric CMOS inverter designed in the previous section. The input was a pulse wave with the following parameters:

- $V1 = 0V$
- $V2 = 1.8V$
- Rise time = 0.1ns
- Fall time = 0.1ns
- Pulse width = 10ns
- Period = 20ns

A transient analysis without any output capacitance was performed first. The output of this analysis is shown in Figure 5.

However, to account for the actual performance of such a circuit, a $10fF$ capacitor was added to the output node. The output of this analysis is shown in Figure 6.

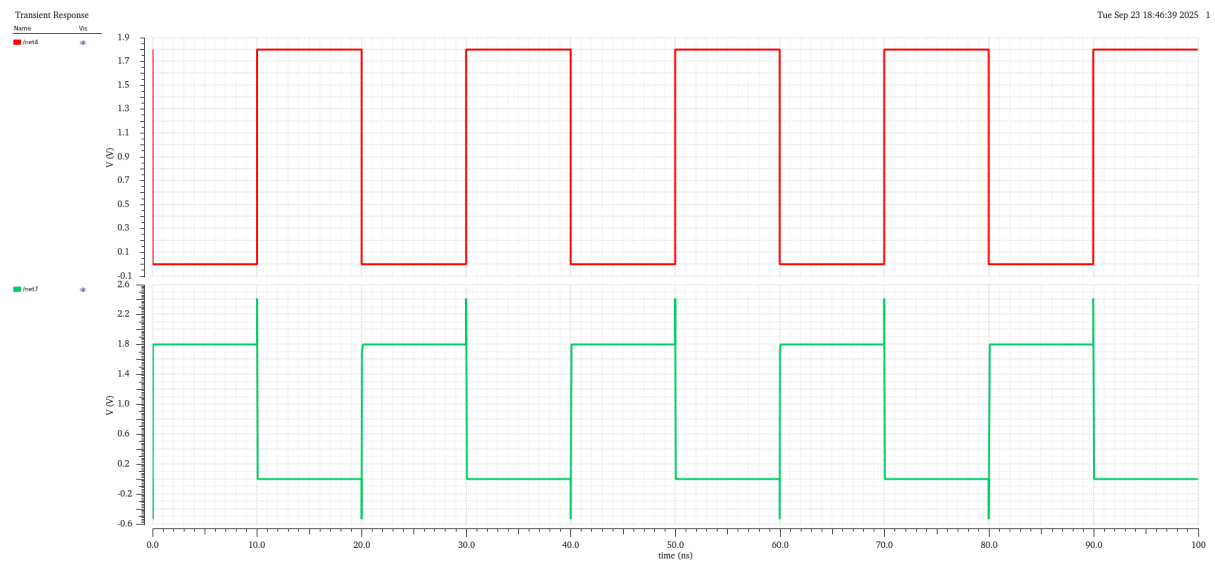


Figure 5: Transient Analysis without Output Capacitance

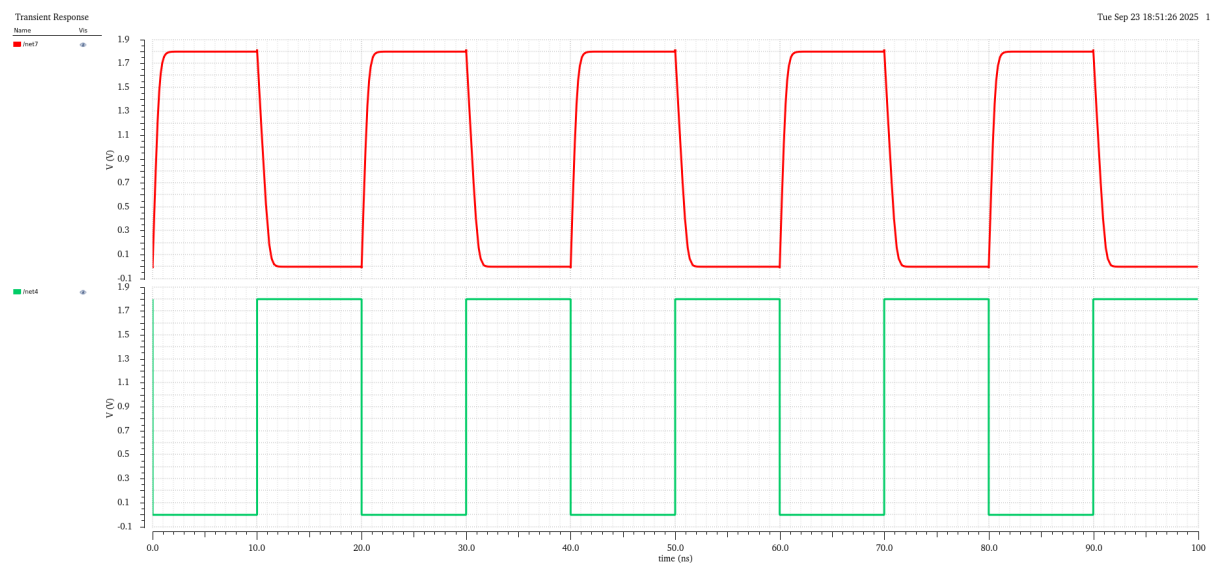


Figure 6: Transient Analysis with 10fF Output Capacitance